# Stop Interpolation Topologically

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# Agenda

Introduction

Persistent Homology

Triangulation of Manifolds

Persistence Stability

Natural Neighbor Interpolation

The Collapsing Sequence

Hypothesis Testing

Summary

# Introduction

# Hypothesis

## Two main assumptions

**Hypothesis 1: Manifold assumption**

A point set $X$ lies on a *topological* or even *smooth manifold* $\mathbb{M}$, having a family of continuous/smooth *coordinate systems* to describe it, with $\dim \mathbb{M} \ll \dim X$.

**Hypothesis 2: Meaning of shape**

The topology of the sublevel sets of a manifold $\mathbb{M}$ underlying a data set can be used to distinguish data up to some equivalence relation.

# Persistent Homology

# An outline to persistent homology

## A brief description

Persistent homology is the study of $k$-**dimensional abelian groups** attached to a topological space. They are studied algebraically and behave *similar* to the topological space itself under change revealing properties of the original space they are attached to by their *invariants*.

The connection of abelian groups by a boundary homomorphism allows to investigate all $k$-dimensions at once. The induced object is a chain complex.

# Chain complex

## How does it work?

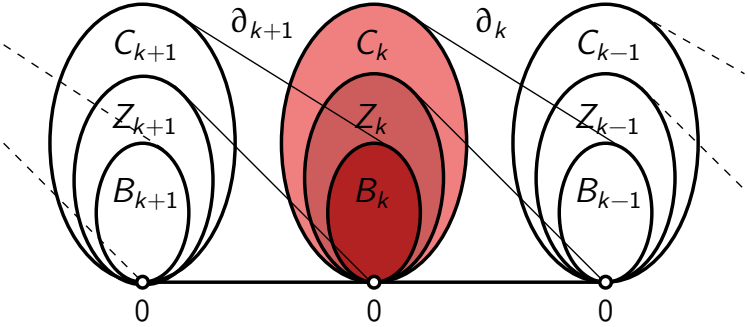
Homology theory is the study of properties of an object $X$, such as a set of points, by means of homological (commutative) algebra. One assigns to $X$ a sequence of abelian groups/modules $C_0, C_1, \cdots$ which are connected by homomorphisms $\partial_{k+1} : C_{k+1} \longrightarrow C_k$ such that $\mathrm{im}\partial_{k+1} \subseteq \ker \partial_k$. Such a structure is called **chain complex**.

$$0 \xrightarrow{\partial_{k+1}} C_k \xrightarrow{\partial_k} C_{k-1} \xrightarrow{\partial_{k-1}} C_{k-2} \xrightarrow{\partial_{k-2}} \cdots \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} 0$$

# Cycle and boundary groups

# Putting them into context

## Illustration of the chain complex




Following Zomorodian, Edelsbrunner and many more ...

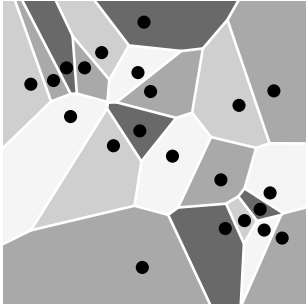
# Triangulation of Manifolds

# Persistence Stability

# Natural Neighbor Interpolation

# Step 1: Voronoi diagram

## Natural neighbor interpolation example

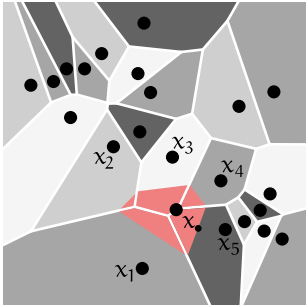


The **Voronoi Diagram** is spanned over the point cloud embedded into some Euclidean space. Technically one does not calculate the Voronoi diagram but the dual **Delaunay triangulation**.

$$\text{Vordgm} = \{x \in X \mid d(x, \lambda_i) \leq d(x, \lambda_j)\} \quad (1)$$

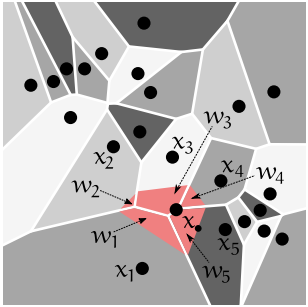
# Step 2: Add a point into the convex hull

## We need to compute the weights per region



# An outline to persistent homology

## A brief description



# An outline to persistent homology

## A brief description

# The Collapsing Sequence

# Hypothesis Testing

# Summary

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